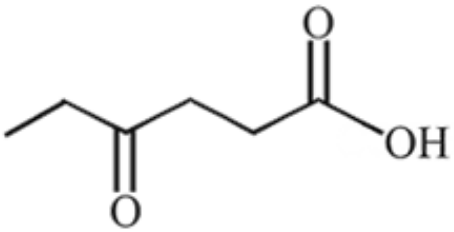
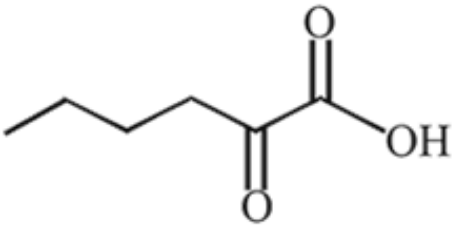
Q.1

Which one of the following compounds undergoes decarboxylation upon heating ?

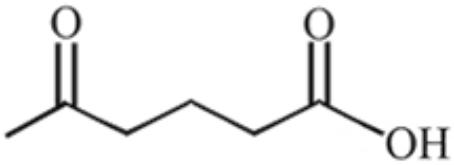
(A)



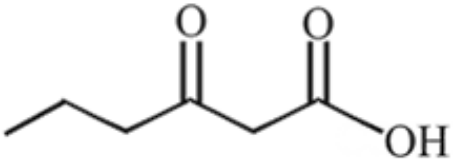
(B)



(C)

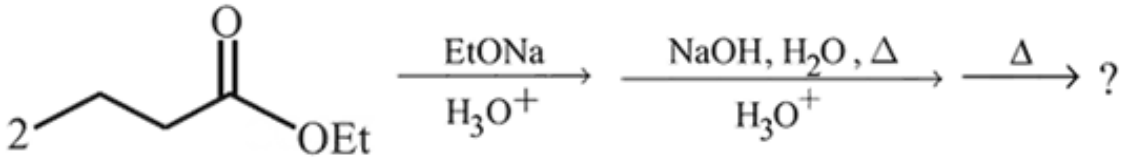


(D)



Q.2

The correct product of the following sequence of reactions



(A) 4 - heptanone

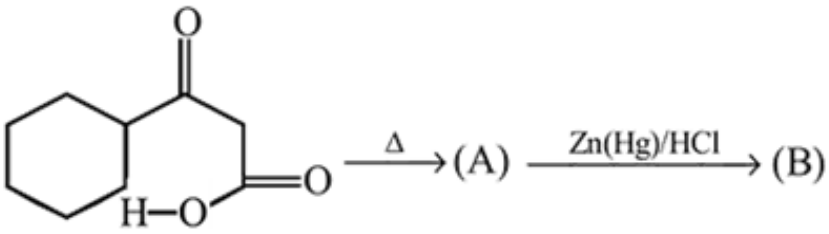
(B) 4— methyl —3 - hexanone

(C) 2 - ethyl pentanoic acid

(D) 2 - propyl butanoic acid

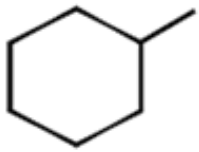
Q.3

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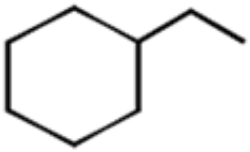


In the above reaction, product (B) is:

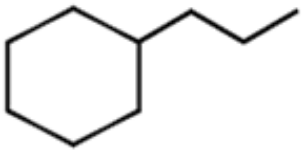
(A)



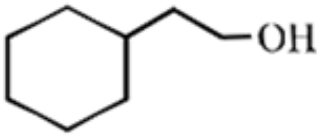
(B)



(C)



(D)



Q.4

$$CH_3CH_2COOH \xrightarrow[\text{Red P}]{Br_2} X \xrightarrow{NH_3\ (alc.)} Y$$

Y in the above reactions is

(A) lactic acid

(B) ethylamine

(C) propylamine

(D) alanine

Q.5

Hydrolsis of an ester gives a carboxylic acid which on Kolbe's electrolysis yields ethane. The ester is

(A) ethyl methanoate

(B) methyl ethanoate

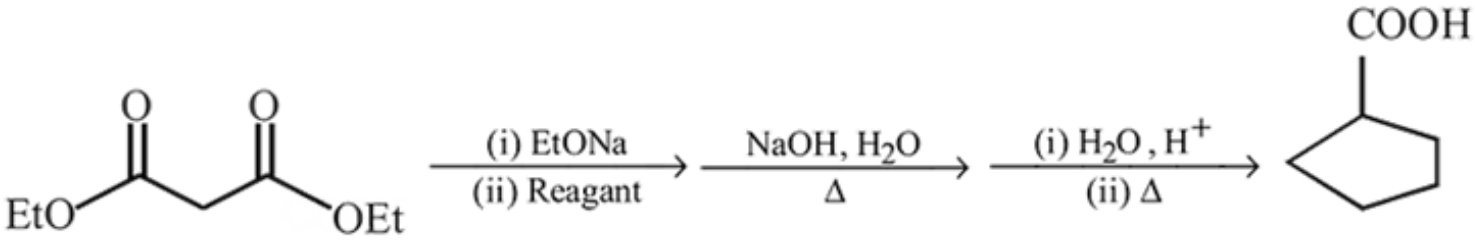
(C) methyl methanoate

(D) methyl propanoate

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Q.6

What is the missing reagent in the synthesis shown below



(A) bromocyclopentane

(B) 1,5 - dibromo pentane

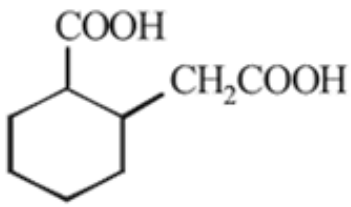
(C) 1,4 - dibromo butane

(D) 1,1 - dibromo cyclopentane

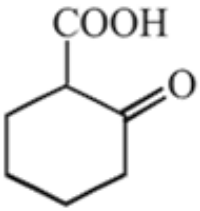
Q.7

The compound that undergoes decarboxylation most readily under mild condition is

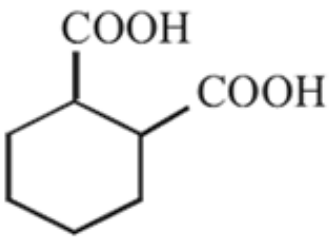
(A)



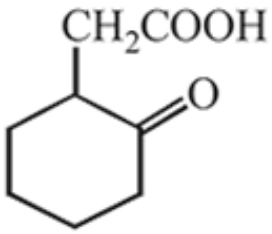
(B)



(C)



(D)



Q.8

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The Hell-Volhard-Zelinsky reaction is used in the synthesis of

(A) aldehydes

(B) α -haloacids

(C) acid halides

(D) ketones

Q.9

The treatment of CH_3CH_2COOH with chlorine in the presence of phosphorus gives

(A) CH_3CH_2COCl

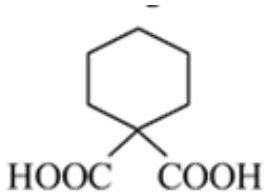
(B) $CH_3CH_2CH_2Cl$

(C) $CH_3CH(Cl)COOH$

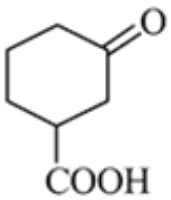
(D) $H_2(Cl)CH_2COOH$

Q.10

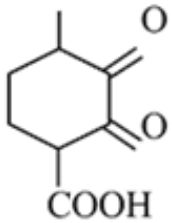
Number of compounds that will undergo decarboxylation on heating



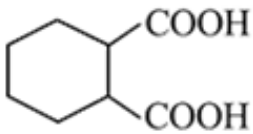
(I)



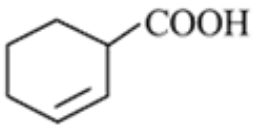
(II)



(III)



(IV)



(V)

(A) 3

(B) 2

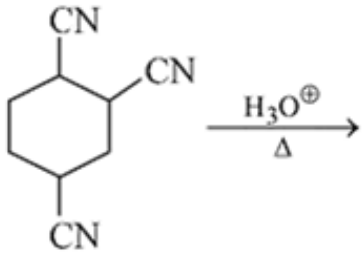
(C) 5

(D) 4

Q.11

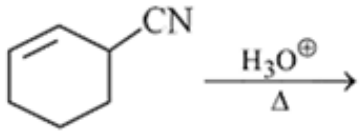
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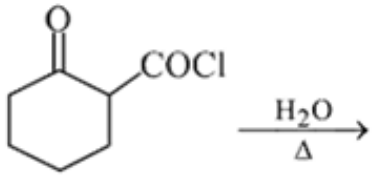
Number of reactions in which product has one carbon lesser than reactant

(I) 

(II) $CH_3CONH_2 + OBr^- \xrightarrow[\Delta]{H_3O^{\oplus}}$

(III) $CH_3COOH + N_3H + H_2SO_4 \xrightarrow{\Delta}$

(IV) 

(V) 

(A) 2

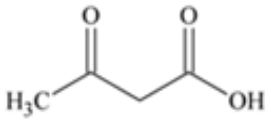
(B) 3

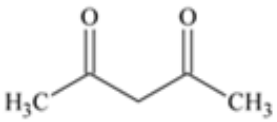
(C) 1

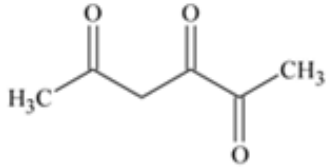
(D) 4

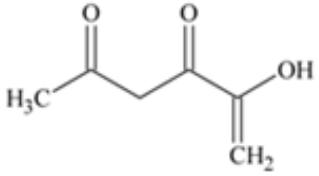
Q.12

Which of the following compound would be expected to decarboxylates when heated

(A) 

(B) 

(C) 

(D) 

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Q.13

Which of the following does not undergo Hell - Volhard Zelinsky reaction?

(A) $HCOOH$

(B) CCl_3COOH

(C) C_6H_5COOH

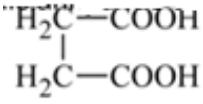
(D) All

Q.14

Which of the following dicarboxylic acids gives a cyclic.anhvdride on heating?

(A) $CH_2(COOH)_2$

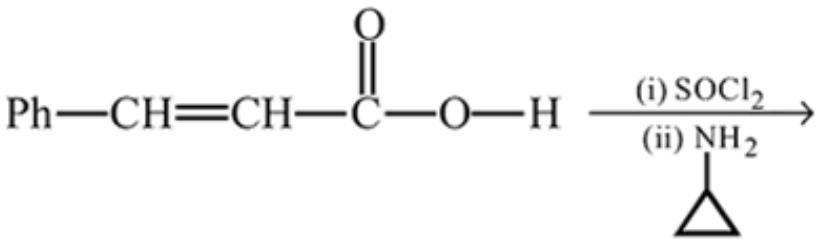
(B)



(C) $HOOC(CH_2)_4COOH$

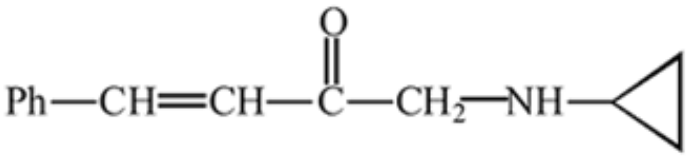
(D) $HOOC(CH_2)_5COOH$

Q.15

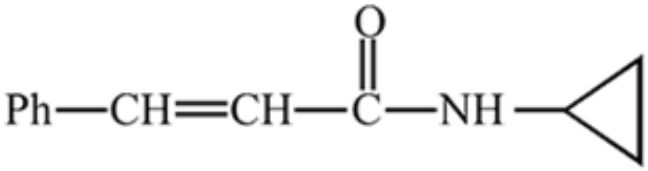


(A); Product (A) of the reaction is:

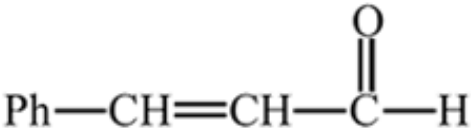
(A)



(B)

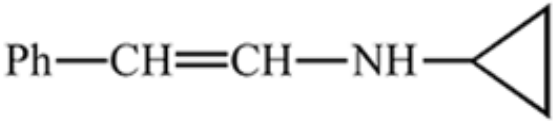


(C)



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(D)



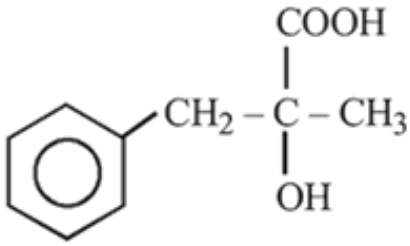
Q.16

In a set of reactions, acetic acid yielded a product D.

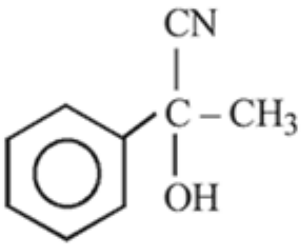
$$CH_3COOH \xrightarrow{SOCl_2} (A) \xrightarrow[AlCl_3]{Benzene} (B) \xrightarrow{HCN} (C) \xrightarrow{HOH} (D)$$

The structure of (D) would be —

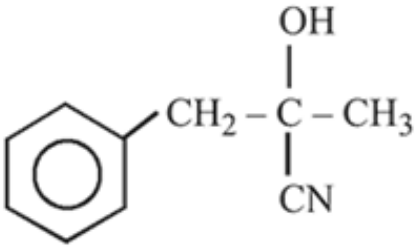
(A)



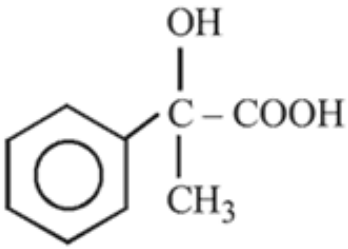
(B)



(C)

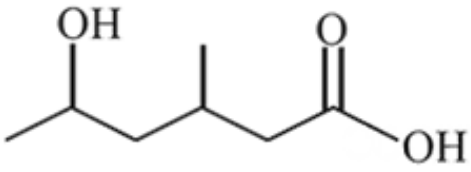


(D)



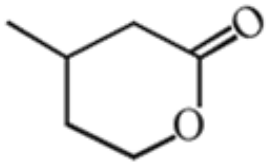
Q.17

Which lactone is formed by heating the following hydroxy acid?

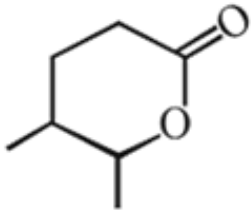


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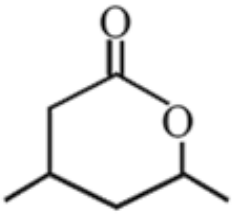
(A)



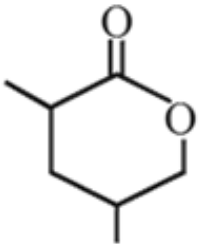
(B)



(C)



(D)



Q.18

Acetic acid on heating in presence of P_2O_5 gives

(A) acetic anhydride

(B) acetylene

(C) peracid

(D) no reaction

Q.19

When acetic acid reacts with ketene, product formed

(A) ethyl acetate

(B) aceto - acetic ester

(C) acetic anhydride

(D) no reaction

Q.20

When propionic acid is treated with aqueous sodium bicarbonate, CO_2 is liberated. The C from CO_2 comes from

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(A) methyl group

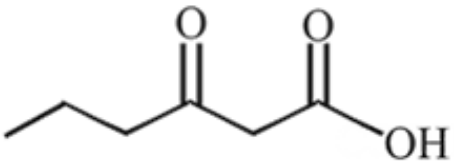
(B) carboxylic acid group

(C) methylene group

(D) bicarbonate

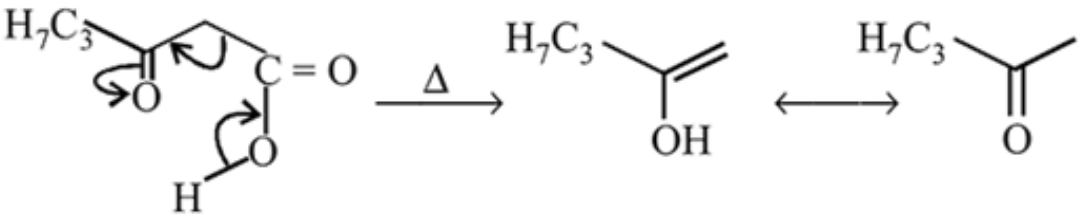
Answers & Solutions

Q.1 Answer:



Solution:

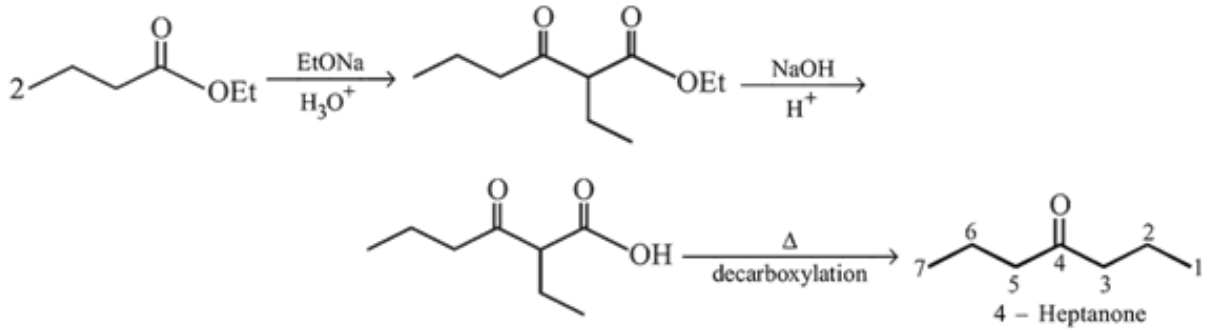
β - Keto acids are decarboxylated by following mechanism



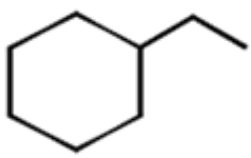
Q.2 Answer:

4 - heptanone

Solution:



Q.3 Answer:



Solution:

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Q.4 Answer:

alanine

Solution:

$$CH_3CH_2COOH \xrightarrow{redP,Br_2} CH_3CHBrCOOH \xrightarrow{NH_3} \underset{\alpha-\text{ Aminopropionic acid (alanine)}}{CH_3CHNH_2COOH}$$

Q.5 Answer:

methyl ethanoate

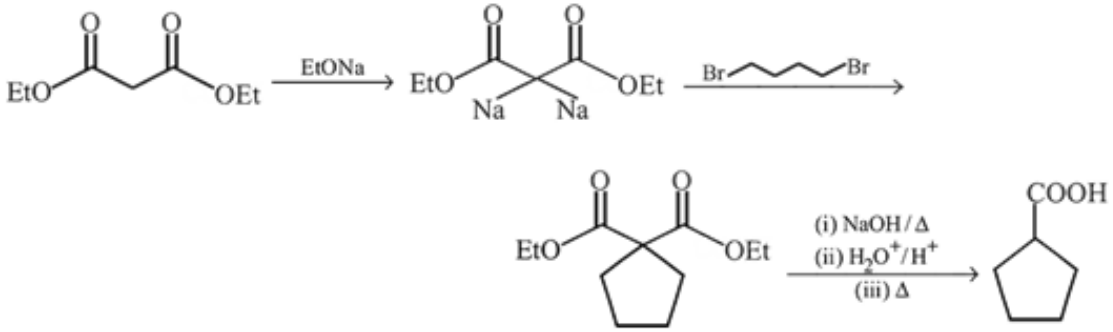
Solution:

Ethane is obtained by the electrolysis of sodium or potassium salt of ethanoic acid (CH_3COOH), hence the ester must be ethanoate CH_3COOR.

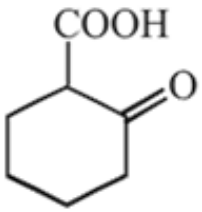
Q.6 Answer:

1,4 - dibromo butane

Solution:



Q.7 Answer:



Solution:

β -Ketoacids undergo decarboxylation easily.

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Q.8 Answer:

α -haloacids

Solution:

The Hell-Volhard-Zelinsky reaction is used in the synthesis of α -haloacids.

Q.9 Answer:

$CH_3CH(Cl)COOH$

Solution:

α -Hydrogen is replaced by chlorine. The product is $CH_3CH(Cl)COOH$

Q.10 Answer:

3

Solution:

(I), (III) and (V) will under go decarboxylation on heating.

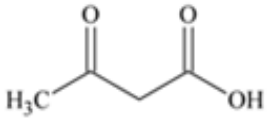
Q.11 Answer:

4

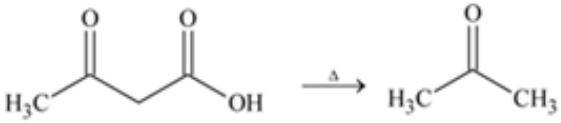
Solution:

II, III, IV and V

Q.12 Answer:



Solution:



In case of b-keto acid, the readly decarboxylation occurs due to 6-membered low energy transition state formation.

Q.13 Answer:

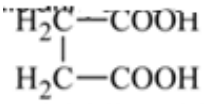
All

Solution:

None of these contains alpha H atom

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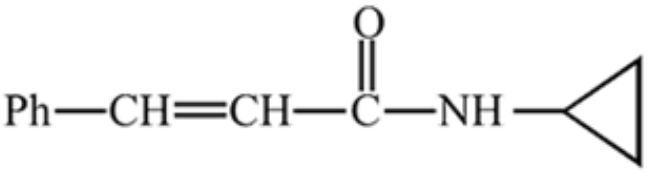
Q.14 Answer:



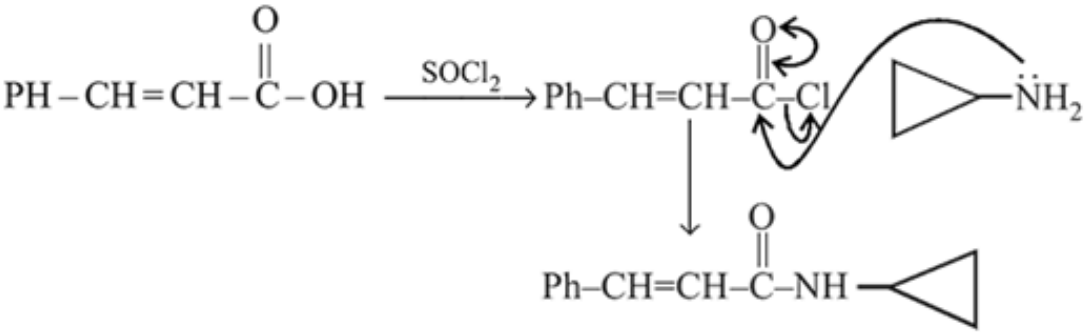
Solution:

Succinic acid on heating forms its anhydride. Option 1, 3&4 are unstable. Only option 2 is stable.

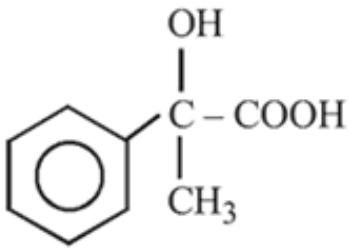
Q.15 Answer:



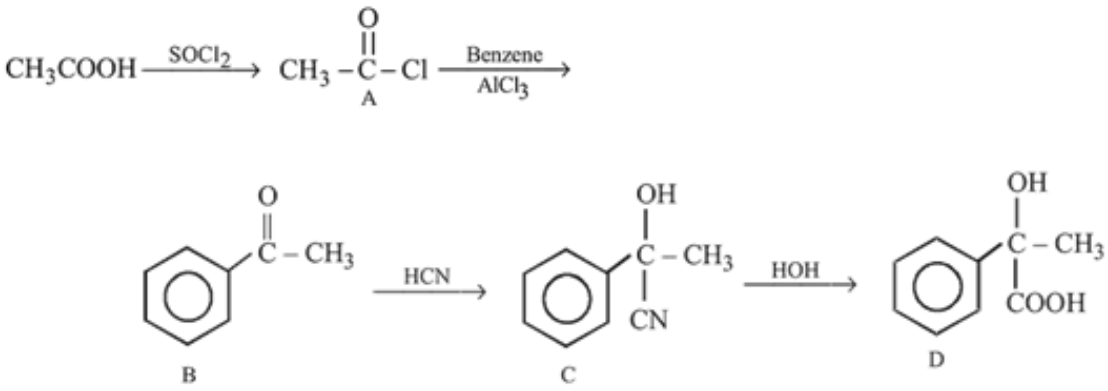
Solution:



Q.16 Answer:

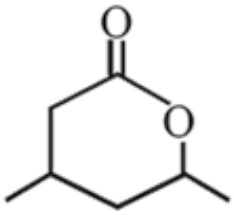


Solution:

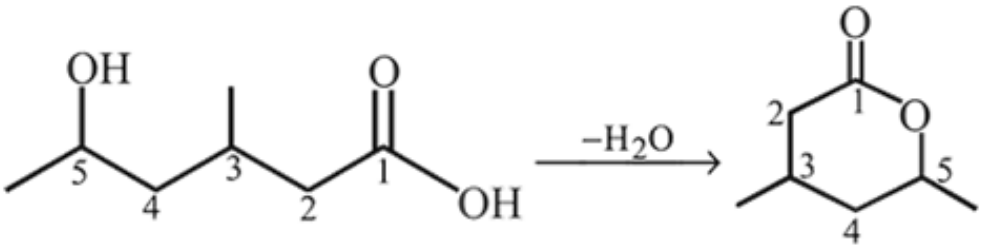


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Q.17 Answer:



Solution:



Q.18 Answer:

acetic anhydride

Solution:

P_2O_5 **is a dehydrating agent and removes a molecule of water from two molecules of acetic acid**

$$CH_3COOH + HOOCCH_3 \xrightarrow{P_2O_5} CH_3COOCOCH_3$$

Q.19 Answer:

acetic anhydride

Solution:

$$CH_3COOH + C_3O_2 \longrightarrow \text{acetic anhydride}$$

(Ketene)

Q.20 Answer:

bicarbonate

Solution:

CO_2 **comes from bicarbonate.**

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